

Program : Diploma in Tool & Die Engineering	
Course Code : 4112	Course Title: Press Tool Technology
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- Give the importance of press tools and understand various die operations
- Select proper tool and a die for a given manufacturing operation to achieve highest productivity
- Introduce the concepts, principles and operations of bending and forming
- Introduce the concepts, principles and operations of drawing and fine blanking

Course Prerequisites:

Topic	Course code	Course name	Semester
Production Process		Production engineering I	2
Production Process		Production engineering II	3
Basics of Moulds and dies Technology		Moulds and dies technology	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the sheet metal processing using different types of press tools	15	Applying
CO2	Describe the Punches, dies and Stoppers for various press operations	14	Understanding
CO3	Illustrate the bending operation and forming operation.	15	Applying

CO4	Illustrate the drawing and fine blanking operations	14	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	2						
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the sheet metal processing using different types of press tools.		
M1.01	Describe the working of presses and parts of press tools.	3	Understanding
M1.02	Explain different press tool operations like blanking piercing, notching, lancing, drawing, bending, forming.	4	Understanding
M1.03	Apply the Importance of clearance in punches and dies operation.	4	Applying
M1.04	Apply the Importance of strip layout and percentage utilization of strip.	4	Applying

Contents:

Press machine - Parts, Specification, classification, Theory of shearing - Shearing action on metal, Plastic deformation, Penetration, Fracture

Die Operations -Blanking, piercing, shearing, cropping, notching, lancing, coining, embossing, stamping, curling, drawing, bending, forming

Cutting clearance - Necessity of clearance, Calculation of clearance, Effects of insufficient, sufficient and optimum clearance, Misalignment between punch and die, Land and angular clearance

Strip layout - narrow run-wide run, single row – double row, Economy factor, calculation of percentage utilization of strip

Cutting force - Calculation of cutting force, methods of reducing the cutting force punch and die, basic requirement of a die.

CO2	Describe the Punches, dies and Stoppers for various press operations.		
M2.01	Distinguish the different types of dies and construction of various dies.	3	Understanding
M2.02	Distinguish different punches and die mountings.	4	Understanding
M2.03	Summarize the knowledge about gauges.	4	Understanding
M2.04	Identify the Fasteners and dowels	3	Understanding
	Series Test I	1	

Contents:

Types of dies and construction-Simple Die, Compound Die, Progressive Die, Combination Die.

Punch & Die mountings- Pilots, Purpose, indirect and direct pilot, size, length of pilot, Strippers- function, types-fixed and traveling stripper, constructional details, spring and fixed stripper, Stripping force for blanking & piercing , Pressure Pads, Knock outs – function, direct & indirect knockouts, stock guide, Stoppers – functions , stop categories, primary, secondary, final, Construction – pin stop, headed pin stop , disappearing pin stop , finger stop, pusher stop , trigger stop, guide bush, guide pins, Side cutters - function – advantages.

Gauges: Functions, dimensional aspect of gauge, Types – back, front, bulge clearance extended back gauge.

Fasteners and dowels: Dowel, function, types, Fasteners – screws, function, types of screws like socket head, counter sunk, cheese head, set screw fasteners.

Types of punches - Classification of punches, Mounting of punches, Materials for punches and dies.

CO3	Illustrate the bending operation and forming operation.		
M3.01	Discuss the theory of bending and various types of bend elements.	3	Understanding
M3.02	Summarize the knowledge about different types of bending operation and spring back phenomenon.	4	Understanding
M3.03	Interpret about the Drawing operation and metal flow during drawing.	4	Applying
M3.04	Prepare the calculation of flat blank diameter and knowledge about types of drawing dies	4	Applying

Contents:

Bending - Theory of bending, Bend elements , Neutral axis and its importance , Blank development calculation -V-bending procedure and force calculation, U-bending theory and

force calculation, Calculation of bend allowance, Bending pressure, Spring back, Methods of avoiding spring back in U and V bending, Bending tool – constructional features, Defects and remedies of bent component and bending tool, Effect of grain direction in bending, Material for bending,

Drawing - Introduction - Metal flow during drawing, Calculations, flat blank diameter, draw reductions, number of draws required, draw force, Variables that affect metal flow during drawing, Types of draw dies, Deep draw tool, Inverted draw tool.

CO4	Illustrate drawing and fine blanking operations		
M4.01	Discuss the forming operation and knowledge about different types of forming operations.	3	Understanding
M4.02	Illustrate the Drawing and deep drawing operations	4	Applying
M4.03	Summarize the Fine blanking operation and operational sequence of fine blanking	4	Understanding
M4.04	Explain the Industrial application of fine blanking and materials suitable for fine blanking	3	Understanding
	Series Test II	1	

Contents:

Forming operations – Introduction, Theory of forming , Types of forming dies, Purpose and application, Different forming operations, Embossing, coining, curling, flanging, planishing, Drawing and deep drawing, cylindrical cup drawing, Wrinkling and puckering,

Fine blanking – Introduction, Construction, Functions, Operational sequences, Applications and advantages, Press selection for fine blanking, Materials suitable for fine blanking work.

Text/ Reference:

T/R	Book Title/Author
T1	Tool design –Cyril Donaldson, Tata McGraw Hill Education Private Limited, New Delhi
R1	Tool and die maker by Mahajan
R2	Design fundamentals by Pacquin
R3	Basic die making - Ostergarrd
R4	Advanced die making - Ostergarrd
R5	Tool design - C.B. Cole
R6	A Textbook of Machine Tool & Tool Design - P. C. Sharma

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/112107144/
2	https://nptel.ac.in/content/storage2/courses/112106153/Module%201/Lecture%201/Lecture_1.pdf
3	http://staff.uny.ac.id/sites/default/files/pendidikan/aan-ardian-mpd/1g-handbook-die-design-2nd-edition.pdf
4	https://nptel.ac.in/courses/112105233/
5	https://nptel.ac.in/courses/112106153/